

CULVERTCRAWL · DESIGN DECISIONS REGISTER

CulvertCrawl design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/culvertcrawl		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

CulvertCrawl design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design-for-construction changes P1 to P14	Accept all; accept some and revisit others	Accept all	The whole build plan follows them	CVC-DDR-003, Table 1
2	Removable laser boom for tight or bent pipes (it overhangs the tracks by about 300 mm)	(a) two bolts at the fin bracket, as modelled; (b) a tool-free clamp	(a) for the prototype; decide (b) after the first pipe trials	Fin bracket (build plan section 3.9)	CVC-DDR-001 item 11; CVC-DDR-002 item 6; CVC-DDR-003, A2
3	Accept the fin's shadow at the invert (6.4 % of the ring in 300 mm pipe)	(a) accept; (b) move the fin off the vertical	(a)	Fin and bracket	CVC-DDR-003, A3
4	Accept the 0.86 kg heavier crawler	(a) accept; (b) thin the ballast plate to 12 mm	(a)	Ballast plate	CVC-DDR-003, A4
5	Which output users need first: a PACP-style observation log, FHWA condition ratings or a deflection report	Any one of the three first	None made	Software only; not part of the TRL 3 build	CVC-DDR-001 item 9; CVC-DDR-002 item 4
6	First partner user group for field trials	County road department, watershed group or university transportation program	None made	Not part of the TRL 3 build	CVC-DDR-001 item 10; CVC-DDR-002 item 5
7	Appearance model differences from <code>model.py</code> (lid status light, render layout of the control case)	Keep for the renders only, or add to the model	Status light: renders only; case position: render layout only	None	Review note, 2026-09-26, items 3 and 5

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The worm gear motor's output shaft is 8 mm D and at least 38 mm long, its output face has two tapped holes, and the shaft sits about 8 mm above the gearbox base and 14 mm from its back	They set the drive pad, the seal counterbore and the side plate holes	CVC-DDR-003, P3
2	The belt kit runs at 200 mm sprocket centres with 36 mm belts, and its sprocket can be bored 8 mm D with a set screw	There is no tensioner; the centres set the belt tension	CVC-DDR-003, P2, P11
3	The dome's flange is 72 mm across and 5 mm thick	It sets the bezel recess and the O-ring groove	CVC-DDR-003, P6
4	The camera board's hole pattern and the lens's position on its thread	They set the camera mount holes and put the lens centre at the dome centre	CVC-DDR-003, P5
5	The laser module is 12 mm across and the cone mirror's base is 20 mm or less	They set the diode housing bore and fit inside the window	CVC-DDR-003, P8
6	The slip ring has an anti-rotation tab or lead the anchor can catch, and passes 100 Mbit/s Ethernet	It sets the anchor bracket; the data link depends on it	CVC-DDR-003, P12
7	The rugged case's inside size and moulded ribs	The chassis base board and panel must drop in with about 2 mm to spare	CVC-DDR-003, P14
8	The battery's size laid on its side (181 x 167 x 77 mm in the model) and that its maker allows it on its side	It sets the battery bay	CVC-DDR-003, P14
9	The tether's aramid member can be broken out and tied off at the eye bolt, and its outside diameter suits the M10 penetrator	Recovery pulls go through it	CVC-DDR-003, P10

■ Value engineering

Value-engineering target: USD 910 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 1,034 (USD 124 over the target). The estimate was USD 906 before the design for construction.

Main cost drivers (the parts added to make the design buildable): the reel frame, bearings, hubs and counter reader (USD 43), the front bezel and the side plates and small made parts (USD 32), the surface box chassis (USD 20), the eye bolt and screws (USD 10), the hull rim and pads (USD 8), the laser fin, window and cap (USD 8), the larger ballast plate (USD 4) and the deck (USD 3). Prices are indicative; the softest (hull, tether, cone mirror) can still move the total by USD 20 to USD 80 either way.

Savings worth trying: a bought cable reel, a printed bezel and a smaller case, then real quotes for the soft prices.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items 1 to 8: budget \$900, pipe range 300 to 900 mm, 12.8 V LiFePO4 surface battery, tracks, 48 V surface power, Class 2 laser ring and fisheye camera, processing on the laptop, pitch unchanged	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	CVC-DDR-001
2026-09-25	Heavier ballast plate for reach (R2), with upstream entry in flooded pipes as operating guidance; ring plane moved to 300 mm ahead (R5); the \$1 overrun accepted	Amish: "i accept all your recommendations, go with them across all repos."	CVC-DDR-002
2026-09-26	Value-engineering target set to \$910 to match the then priced BOM (\$906)	Amish: "i approve all the budget items."	CVC-DDR-002 v0.2
2026-09-30	Make the design physically buildable while drawing the build plan; changes recorded in CVC-DDR-003 and open for review (open decision 1)	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	CVC-DDR-003
2026-09-30	Outstanding decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	STANDARDS section 18